

# 工作坊 2：教師準備及流程

## Workshop 2: preparation and flow

每堂 60 分鐘；兩人一組。目標是執行、改數據、解釋預測。

60 minutes, pairs. Aim to run code, change data and explain predictions.

課前：確認學校 Google 帳戶可用 Colab；檢查網絡及鍵盤裝置。

Before class: check approved Google accounts, Colab access, internet and keyboards.

資源：學生 Notebook 的結果已清空，以保護「先猜，再看結果」。

The student notebook has cleared outputs so students can guess before the reveal.

程式已在本機逐格執行檢查；仍需老師在實際課室裝置試開 Colab。

Code was executed locally; the teacher still needs to rehearse on classroom devices.

0–8 分鐘：老師示範打招呼，學生猜結果及提出新訊息。

0–8: teacher greeting demo; students guess and suggest messages.

8–20 分鐘：開連結、儲存副本、再做熱身及準備工具。

8–20: open the link, save a copy, repeat the warm-up and set up tools.

20–35 分鐘：散點先猜；再執行「3 建立模型」和「4 作預測」。

20–35: guess from dots, then run named sections 3 Fit and 4 Predict.

35–50 分鐘：先猜 (4,95) 的影響，再跑比較；每次只改一處。

35–50: predict the effect of (4,95), run the comparison, change one thing.

50–60 分鐘：比較 3.5/10 小時，寫觀察，分享一個發現。

50–60: compare 3.5/10 hours, write observations and share a discovery.

參考：3.5 小時約 57 分；10 小時約 70 分；加 (4,95) 後約 62 分。

Reference: 3.5 h  $\approx$  57; 10 h  $\approx$  70; after adding (4,95), 3.5 h  $\approx$  62.

接受近似讀值及合理描述；不要求計算 m、c 或解釋每行程式。

Accept approximate readings and descriptions; no hand calculation or line-by-line mastery.

遇錯先看提示。改原數據後，重跑「2 載入數據」及「3 建立模型」。

After editing original data, rerun the named Load data and Fit model sections.

模型輸出有變化，不代表真實準確率有提升；成績數據皆為模擬。

A changed output is not proof of better real-world accuracy. Scores are simulated.

若學生不能登入，兩人共用可用帳戶／裝置，或用老師示範及工作紙。

If access fails, pair on an approved working device or use the teacher demo and worksheet.

Resources: [gordany.github.io/f2-linear-regression/workshop-2/](https://gordany.github.io/f2-linear-regression/workshop-2/)

# 工作坊 3：教師準備及流程

## Workshop 3: preparation and flow

課前：準備筆、擦膠、空背景；先試相機權限及 Teachable Machine。

Before class: prepare pens, erasers and empty backgrounds; test camera access and the site.

由老師預先拍攝 24 張測試相片，分 A/B，均不放入訓練圖片。

Teacher prepares 24 test photos in A/B folders, kept separate from all training images.

每批：筆 4 張、擦膠 4 張、只有背景 4 張；B 用新的拍攝，不是複製 A。

Each set: 4 pen, 4 eraser, 4 background-only photos. B uses fresh shots, not copies of A.

A/B 採相近條件：正面、斜角、較遠、新背景；背景類須沒有物件。

Use similar conditions in A/B: front, angled, farther away and another background.

檔案名例如 A/pen\_01.jpg、B/pen\_01.jpg；另有 eraser\_01、empty\_01 至 04。

Name files by set/class, e.g. A/pen\_01.jpg and B/pen\_01.jpg; number each class 01–04.

派 A/B 資料夾給每組。B 保持未用，改善後才以 Preview 的圖片輸入測試。

Distribute A/B folders. Keep B unused until final testing via Preview's image input.

0–8 分鐘：藏物件／換背景小遊戲，先寫預測。

0–8: hidden-object/background game and written guesses.

8–20 分鐘：圖像、CNN 的想法，建立三類，解釋預留 A/B。

8–20: image/CNN ideas, set up three classes and explain A/B.

20–35 分鐘：各類目標 20 張以上有變化的訓練圖片，再訓練。

20–35: aim for 20+ varied training images per class, then train.

35–55 分鐘：測 A、補充例子、重訓、測未用 B；每類各記 4 張。

35–55: test A, add examples, retrain, test fresh B; record counts out of 4.

55–60 分鐘：反思模型分數、新圖片測試及未教物件。

55–60: reflect on model scores, new-image testing and unfamiliar objects.

A 全部答對可寫「未見錯誤」。不要造錯誤，也不要提早用 B。

If A is all correct, write “none found”. Do not invent errors or use B early.

20 張是課堂目標，不是產品門檻。模型分數不等於測試準確率。

20 images is a classroom target, not a product minimum. Scores are not test accuracy.

借用預訓練圖像特徵，建立自己的分類部分；不需從零訓練整個 CNN。

Reuse pretrained image features and teach the classifier; no full CNN training from scratch.

只拍物件。若相機不可用，用老師備妥的訓練相片；A/B 仍須分開。

Use object photos. If camera access fails, supply training photos while keeping A/B separate.

Resources: [gordany.github.io/f2-linear-regression/workshop-3/](https://gordany.github.io/f2-linear-regression/workshop-3/)